TECHNICAL MEMORANDUM

BBMI NCAA Baseball Model

TO:	BBMI Sports Analytics — Internal Technical Record
FROM:	BBMI Quantitative Modeling
DATE:	April 1, 2026
RE:	NCAA Baseball Model — Technical Memorandum
MODEL VERSION:	V4 (Poisson Engine, ERA-Based Pitcher Adjustment, Self-Computed Park Factors)
VALIDATION PERIOD:	2024–2025 NCAA Seasons (Walk-Forward)

This memorandum contains proprietary modeling methodology and validation results for the BBMI NCAA Baseball prediction system. Results are based on walk-forward out-of-sample testing. Past performance does not guarantee future results.

1. Executive Summary

The BBMI NCAA Baseball model projects game-level scoring using a Poisson engine with two core team ratings (offense and defense), pitcher-specific adjustments, park factors, and conference tier offsets. The model produces two validated wagering products:

Product	Walk-Forward ATS	Sample	Status
Total Under (edge ≥ 1.5 runs)	**61.7%**	436 games (2025)	Validated — primary product
Spread (edge 2.0–5.0 runs)	**54.4%**	1,977 games	Validated — co-primary

The model is structurally an under-finder — it identifies games where the projected total is meaningfully below the Vegas posted line. The under product achieves a higher ATS than the spread product because college baseball totals are less efficiently priced by the market than spreads.

2. Model Architecture

The model pipeline executes the following sequential steps:

```
Step 1    Calculate team offensive rating (wOBA, OBP, SLG, SB, SOS, adj R/G)
Step 2    Calculate team defensive rating (FIP, bullpen ERA, WHIP, K/9)
Step 3    Apply re-anchoring (force league-average ratings to match actual league R/G)
Step 4    Base run projection = (team_off + opponent_def) / 2
Step 5    Apply pitcher adjustment (ERA-based, Bayesian shrinkage)
Step 6    Apply park factor
Step 7    Apply home-field advantage (+0.6 runs total)
Step 8    Apply conference tier offset
Step 9    Apply series position and day-of-week adjustments
Step 10   Apply workload and bullpen fatigue adjustments
Step 11   Poisson win probability from projected runs
Step 12   Spread = margin × BBMI_LINE_MULTIPLIER (2.0)
Step 13   Total = home_proj + away_proj + day-of-week adjustment
Step 14   Edge calculation vs Vegas
```

Design Principles

- Team ratings, not individual matchups. The model computes team-level offensive and defensive ratings. Individual pitcher adjustments are layered on top but cannot override the team signal. Walk-forward testing confirmed team FIP captures 85% of the pitcher signal.
- Poisson scoring. College baseball games have lower, more discrete scoring patterns than MLB. The Poisson distribution is an adequate approximation for college baseball where fewer extreme blowouts occur.
- Re-anchoring. Offensive and defensive ratings are dynamically re-anchored so that the league-average composite always equals the actual observed league-average R/G. This prevents systematic over- or under-projection as the scoring environment shifts during the season.
- Under-first product hierarchy. The model's highest-validated signal is in identifying unders. Over picks (44% ATS) are an anti-signal. The model is calibrated to maximize under pick quality.

3. Input Data and Calibration

3.1 Team Offensive Rating

Six components are weighted to produce an offensive runs-per-game estimate:

Component	Weight	Scaling	Description
adj_runs_per_game	0.25	Raw	Adjusted runs scored per game
wOBA (or OPS)	0.25	× 15.0	Weighted on-base average
OBP	0.20	× 12.0	On-base percentage
SLG	0.10	× 10.0	Slugging percentage
Stolen base rate	0.10	× 2.0	Team speed/aggressiveness
SOS adjustment	0.10	(150 − SOS_rank) / 300	Strength of schedule correction

Regression to mean: 25% blend toward league average (6.5 R/G baseline, dynamically updated). Re-anchoring: the final rating is multiplied by _REANCHOR_OFF (currently 1.308) to force the league-average offensive rating to equal the actual observed league R/G.

3.2 Team Defensive Rating

Five components (one disabled) are weighted to produce a defensive runs-allowed-per-game estimate:

Component	Weight	Description
Team FIP (starters)	0.35	Fielding-independent pitching of starting rotation
Bullpen ERA/FIP	0.25	Relief pitching quality
WHIP	0.20	Walks + hits per inning pitched (× 3.0 scaling)
K/9	0.20	Strikeouts per 9 innings (inverted: (10 − K/9) × 0.5)
Fielding %	0.00	Removed — notoriously poor quality in college data

3.3 FIP Formula

```
FIP = (13 × HR + 3 × (BB + HBP) − 2 × K) / IP + 3.2
```

Key finding: In college baseball, FIP and ERA correlate at r = 0.157 vs r = 0.158. Theory (FIP) does not outperform direct observation (ERA) because noise in the college game swamps the theoretical advantage of FIP's defense-independence. Both are used: FIP for team ratings (more stable), ERA for individual pitcher adjustments (more responsive).

3.4 Pitcher Adjustment (Bayesian Shrinkage)

```
shrunk_ERA = individual_weight × pitcher_ERA + team_weight × team_ERA
adjustment = clamp((shrunk_ERA − team_ERA) × PITCHER_K, −2.0, +2.0)
```

Where PITCHER_K = 0.18 (calibrated on 1,114 games with known starters).

Starts	Individual Weight	Team Weight
0–2	0%	100%
3	37.5%	62.5%
5	62.5%	37.5%
8+	85%	15%

Cap: ±2.0 runs maximum adjustment per pitcher. This prevents a single outlier pitcher ERA from dominating the projection.

3.5 Park Factors

Self-computed from observed data with team-quality adjustment:

```
implied_pf = (team_home_rpg × 2) / (team_home_rpg + team_away_rpg)
```

Adjusted for team quality (strong teams inflate home RPG by winning more), with 30% regression toward 1.0 and weekly updates. Minimum 8 home games required.

Category	Examples	Factor Range
High altitude / hitter-friendly	Colorado State (1.15), Air Force (1.15), Wyoming (1.12)	1.06–1.15
Power conference pitchers' parks	Vanderbilt (0.92), Virginia (0.93), Stanford (0.93)	0.92–0.96
Neutral	Most parks	0.97–1.03

3.6 Home-Field Advantage

HCA = 0.6 runs (flat, applied as +0.3 home / −0.3 away). The HCA adjustment cancels in the total projection and only affects the spread. Team-specific home/away splits were tested and rejected (stability r = 0.026 — pure noise).

4. Projection Engine

4.1 Base Run Projection

```
home_proj = (home_off + away_def) / 2.0
away_proj = (away_off + home_def) / 2.0
```

This crosses each team's offensive capability against the opponent's defensive quality. The average produces a balanced estimate that accounts for both sides of the matchup.

4.2 Sequential Adjustments

Adjustment	Formula	Description
Pitcher	+/− (shrunk ERA − team ERA) × 0.18	Individual starter quality
Park factor	× park_factor	Venue-specific scoring environment
HCA	+0.3 home / −0.3 away	Home-field advantage
Conference tier	+/− offset / 2 per team	Cross-tier matchup adjustment
Series position	Defense × 0.95/1.00/1.04	Fri ace / Sat avg / Sun weaker
Day of week (total)	−2.0 Fri / −1.5 Sat / +0.3 Sun	Starter workload effect
Midweek	× 1.03	Tue/Wed/Thu weaker pitching
Bullpen fatigue	+0.3 if BP 10+ IP in 3 days	Opponent scores more
Workload adj	Based on starter IP vs 4.4 baseline	Reliever innings cost
Platoon	+0.15 if LHP vs RHB-dominant	Handedness advantage

4.3 Worked Example

LSU (home) vs Vanderbilt (away). Friday night. LSU park factor 1.06.

```
LSU OFF = 7.8, LSU DEF = 5.2
Vandy OFF = 6.5, Vandy DEF = 4.1

Base: LSU = (7.8 + 4.1) / 2 = 5.95   |   Vandy = (6.5 + 5.2) / 2 = 5.85
Pitcher (LSU ace ERA 2.80, team 3.90, k=0.18, 6 starts):
  shrunk = 0.70×2.80 + 0.30×3.90 = 3.13  |  adj = (3.13−3.90)×0.18 = −0.14
  Vandy = 5.85 − 0.14 = 5.71
Park factor:   LSU = 5.95×1.06 = 6.31   |   Vandy = 5.71×1.06 = 6.05
HCA:           LSU = 6.31 + 0.30 = 6.61  |   Vandy = 6.05 − 0.30 = 5.75
Series (Fri):  Vandy = 5.75×0.95 = 5.46   |   LSU = 6.61×0.95 = 6.28
DOW total adj: −2.0 (Friday)
BBMI Total = 6.28 + 5.46 − 2.0 = 9.74  → snapped to 9.5
BBMI Spread = (6.28 − 5.46) × 2.0 = 1.64 → LSU −1.5
```

5. Scoring Distribution — Poisson Engine

5.1 Win Probability

```
P(home wins) = Σ[h=0..max, a=0..max] P(h) × P(a) × I(h > a)
             + P(tie) × POISSON_HOME_TIE_SHARE

P(k) = e^−λ × λ^k / k!
POISSON_HOME_TIE_SHARE = 0.53  (extra-inning walk-off advantage)
```

5.2 Moneyline Derivation

```
if win_prob >= 0.50:   ML = −round(win_prob / (1 − win_prob) × 100)
else:                  ML = round((1 − win_prob) / win_prob × 100)
```

5.3 Standard Deviations (Edge Calculation)

Product	STD_DEV	Notes
Spread	6.2	Calibrated from 710 games
Total	4.0	Production value; 2026 data suggests 6.24

6. Product Definitions and Examples

6.1 Total Under (Primary Product)

```
ou_edge = vegas_total − bbmi_total
if ou_edge ≥ 1.5  →  UNDER recommendation
```

Confidence Level	Edge
High	≥ 3.0 runs
Moderate	2.0–3.0 runs
Low	< 2.0 runs

Walk-forward: 61.7% ATS on 436 games (2025). The model's highest-confidence product.

6.2 Spread (Co-Primary Product)

```
spread_edge = |bbmi_line − vegas_line|
if bbmi_line > vegas_line  →  pick away
if bbmi_line < vegas_line  →  pick home
```

Optimal edge range: 2.0–5.0 runs. Walk-forward: 54.4% ATS on 1,977 games (2024–2025 combined). Both seasons independently profitable: 54.1% (2024), 54.6% (2025).

6.3 Over (Display Only — Anti-Signal)

Over picks hit at approximately 44% ATS — worse than a coin flip. Over picks are displayed for transparency but carry an explicit display-only label.

6.4 Model Maturity Labels

Label	V4 Games	Status
Immature	< 500	Early season
Developing	500–1,500	Growing sample
Calibrating	1,500–5,000	Current (2,355 games, Apr 2026)
Mature	5,000+	Full confidence

7. Walk-Forward Validation Protocol

7.1 Methodology

2024 team ratings (built from 2024 season data) are applied to 2025 games as a walk-forward test. The multiplier (2.0) is calibrated from 2024 in-sample data and held fixed for 2025 out-of-sample evaluation.

7.2 Multiplier Calibration

BBMI_LINE_MULTIPLIER = 2.0 (converts raw rating differential to spread scale). Calibration plateau at 1.75–2.25. Center-of-plateau chosen (2.0) for robustness.

Multiplier	ATS
1.75	53.9%
2.00	**54.4%**
2.25	54.1%

8. Validated Results

8.1 Total Under

Metric	Value
ATS (2025 walk-forward)	**61.7% (269/436)**
Edge threshold	≥ 1.5 runs
ROI at −110	**~17.4%**

8.2 Spread

Metric	Value
ATS (combined 2024–2025)	**54.4% (1,077/1,977)**
2024 season	54.1%
2025 season	54.6%
Edge 2.0–5.0 runs	**55.2%**

8.3 Calibration Diagnostics (2026 In-Sample)

Metric	Spread	Total
STD_DEV	5.87	6.24

MAE	4.56	5.03
Bias	−0.348 (slight home under-projection)	−0.734 (under-projecting 0.7 R/G)

9. Components Investigated and Excluded

9.1 PQS (Pitcher Quality Score)

Hypothesis: A theory-based pitcher quality metric combining IP, hits, walks, strikeouts, and home runs would outperform raw ERA for individual pitcher adjustments.

Result: PQS produced r = 0.008 correlation with margin outcomes. Raw ERA produced r = 0.158. PQS destroyed signal in the walk-forward.

Conclusion: Direct observation (ERA) beats theory (PQS) in the high-noise college baseball environment. ERA-based adjustment with k = 0.18 is the validated approach.

9.2 Home/Away Splits

Result: Split stability correlation r = 0.026 (first half to second half). Pure noise. Flat HCA of 0.6 runs captures the advantage. Team-specific splits add error.

9.3 Fielding Percentage

Permanently excluded. Fielding percentage is a notoriously poor quality measure in college baseball — it penalizes teams with more athletic fielders who attempt difficult plays.

9.4 CBI Park Factors (External Source)

CBI stadium RPG had a +1.64 run systematic bias. The values were inflated by team quality (good teams play in parks that appear hitter-friendly because the team scores a lot, not the park). Self-computed park factors with team-quality adjustment, 30% regression, and an 8-game minimum are now used.

9.5 Complex Pitcher Workload Models

Minimal impact after re-anchoring handles the total bias. Capped at 1.5 runs to preserve the under signal. Simplified version retained (SP_RP_ERA_GAP = 2.12, WORKLOAD_ADJ_CAP = 1.5).

9.6 K/9 Weight Swap

Tested: Reducing bullpen ERA weight from 0.25 to 0.15 and increasing K/9 from 0.20 to 0.30.

Result: Spread improved +0.2–0.4 pp, but under degraded −0.6 to −1.2 pp.

Decision: Under product has veto power. Weights locked as-is.

10. Open Items and Forward-Looking Agenda

10.1 Monitoring Thresholds

Metric	Target	Trigger
Under ATS (live 2026)	≥ 55%	Review if < 52% at 500 games
Spread ATS (live 2026)	≥ 53%	Promote to full recommendation at 500 games if > 53%
STD_DEV_TOTAL	4.0 (production)	2026 data suggests 6.24 — recalibrate at season end
Total bias	0.0 target	Current −0.734 under-projection; re-anchoring should correct as season progresses

10.2 Deferred Enhancements

Item	Description	Dependency
Umpire strike zone data	Could improve F/B% and K-rate projections	Data availability
Lineup-specific platoon adj	Currently flat 0.15; could be LHP-matchup-specific	Daily lineup feeds
Weather integration	Temperature/wind effects on scoring	Weather API + park identification
Conference tournament adj	Neutral-site, single-elimination dynamics	Venue + format data

10.3 Weight Lock Policy

Weights are locked and not to be modified unless new orthogonal data sources are added (platoon, umpire, daily lineup). The under product has veto power over any weight change — if a proposed change improves spread ATS but degrades under ATS, the change is rejected.

Appendix A — Rating Component Weights

Offensive Rating

Component	Weight	Scaling
Adjusted R/G	0.25	Raw
wOBA (or OPS)	0.25	× 15.0
OBP	0.20	× 12.0
SLG	0.10	× 10.0
Stolen base rate	0.10	× 2.0
SOS adjustment	0.10	(150 − rank) / 300

Defensive Rating

Component	Weight	Description
Team FIP (starters)	0.35	Rotation quality
Bullpen ERA/FIP	0.25	Relief quality
WHIP	0.20	Baserunner management
K/9	0.20	Strikeout rate (inverted)
Fielding %	0.00	Removed

Appendix B — Conference Tiers

Tier 1 (Power)

SEC, ACC, Big 12, Big Ten, Pac-12

Tier 2 (Mid-Major)

Sun Belt, American, Mountain West, C-USA, MAC, West Coast, Missouri Valley, Colonial

Tier 3 (Low-Major)

All other conferences

Cross-Tier Run Offsets

Matchup	Total Adjustment
Tier 1 vs Tier 1	0.0
Tier 1 vs Tier 2	+0.3

Tier 1 vs Tier 3	+0.6
Tier 2 vs Tier 2	0.0
Tier 2 vs Tier 3	+0.2
Tier 3 vs Tier 3	0.0

Each team receives half of the total adjustment applied to their projected runs.

Appendix C — Glossary

Term	Definition
FIP	Fielding Independent Pitching — measures pitcher quality using only HR, BB, HBP, K
wOBA	Weighted On-Base Average — offensive metric weighting each event by run value
OBP	On-Base Percentage — rate at which batters reach base
SLG	Slugging Percentage — total bases per at-bat
WHIP	Walks + Hits per Inning Pitched
K/9	Strikeouts per 9 innings
SOS	Strength of Schedule — opponent quality ranking
Re-anchoring	Dynamic calibration forcing league-average rating to equal actual league R/G
Bayesian Shrinkage	Blending individual pitcher stats with team average, weighted by sample size
Series Position	Friday (ace), Saturday (middle), Sunday (weaker) — reflects pitching quality in college weekend series
Poisson	Probability distribution modeling discrete events (runs scored) where variance equals the mean
PITCHER_K	Scaling constant (0.18) converting ERA deviation to run adjustment
Park Factor	Self-computed venue adjustment for scoring environment (1.0 = neutral)

This memorandum describes the BBMI NCAA Baseball model as of April 1, 2026. Model parameters and validation results are subject to revision based on ongoing monitoring and end-of-season review.